



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



NASA Core Flight System Vulnerability Daily Space-Systems Review

Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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NASA Core Flight System Vulnerability Daily Space-Systems Review

1. Executive Summary

CISA advisories for NASA Core Flight System Health and Safety application vulnerabilities illustrate why space and aerospace operators should track open-source mission software exposure and denial-of-service conditions in non-production and operational ground environments. This daily Lost Boys Cyber report is intended to convert current-source coverage into immediate analyst action: validate exposure, collect the right telemetry, and run focused hunts before the activity becomes a stale backlog item.

The assessment is TLP:CLEAR and based only on public reporting available for the run date. Claims that are not directly stated by public sources are framed as defender implications rather than confirmed victimology or attribution.

2. Intelligence Requirements

Question	Current Answer	Confidence
Why should defenders review this today?	It reflects a current or recently updated threat pattern with practical exposure in enterprise environments.	Medium
Which teams should act?	SOC, vulnerability management, identity, endpoint, cloud/SaaS, and affected business-system owners.	Medium
What evidence should be collected first?	Endpoint process/network telemetry, identity sign-ins, admin audit logs, proxy/DNS logs, and third-party access logs.	High

3. Source Review & Confidence

#	Source	URL	Contribution
1	CISA NASA cFS ICSA-26-211-06	https://www.cisa.gov/news-events/ics-advisories/icsa-26-211-06	Current CISA ICS advisory for cFS HS incomplete fix/NULL pointer dereference.
2	CISA NASA cFS ICSA-26-197-03	https://www.cisa.gov/news-events/ics-advisories/icsa-26-197-03	Related cFS HS denial-of-service advisory.
3	NASA Core Flight System project	https://cfs.gsfc.nasa.gov/	Project context and software ecosystem reference.

4. Threat Activity Overview

CISA advisories for NASA Core Flight System Health and Safety application vulnerabilities illustrate why space and aerospace operators should track open-source mission software exposure and denial-of-service conditions in non-production and operational ground environments.

Activity Element	Analyst Assessment	Collection Priority
Initial access	Likely to involve exposed services, identity abuse, social engineering, or trusted supplier compromise depending on	High

	environment.	
Execution / foothold	Look for script execution, unusual child processes, package install hooks, RMM/tooling abuse, or browser-mediated staging.	High
Command and control	Monitor direct-to-IP, unusual TLS destinations, and abnormal SaaS/API usage.	Medium
Impact	Data theft, ransomware, service disruption, or supply-chain propagation are plausible business impacts.	Medium

5. Affected Sectors and Exposure

Sector / Environment	Relevance	Priority Checks
Enterprise IT	Identity, endpoints, SaaS, and perimeter services are common attack surfaces.	MFA posture, EDR coverage, admin audit logs
Managed service / supply chain	Trusted relationships amplify blast radius.	Vendor access inventory, RMM restrictions, package provenance
Critical operations	Outages may become operational disruptions even when OT is segmented.	Business continuity playbooks, segmentation, monitoring

6. Aviation and Aerospace Sector Impact

Segment	Operational Relevance	Exposure Path	Defender Action
Airlines / airports	Disruption can affect passenger processing, baggage, crew scheduling, and operations coordination.	Shared SaaS, identity providers, VPN, help desk, managed IT.	Validate third-party access, enforce MFA-resistant controls, rehearse manual fallback.
Aerospace / defense manufacturing	IP theft and production disruption can affect export-controlled design, supplier delivery, and maintenance schedules.	Engineering SaaS, supplier portals, RMM, VPN, code/package dependencies.	Segment engineering networks and monitor privileged vendor access.
Satellite / space / avionics support	Ground systems and telemetry pipelines can be high-impact even when flight safety systems are isolated.	Ground-segment IT, mission operations workstations, CI/CD, vendor appliances.	Review remote access, log retention, and incident escalation with mission teams.
Aviation supply chain / MRO	Third-party compromise can cascade into airline and OEM operations.	Contractor identity, ticketing, document exchange, update mechanisms.	Inventory trust relationships and require breach notification playbooks.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed or Plausible Technique	Evidence / Rationale
Initial Access	T1190 Exploit Public-Facing Application /	Current public reporting emphasizes

	T1566 Phishing / T1195 Supply Chain Compromise	exploitation, social engineering, or package/supplier paths.
Execution	T1059 Command and Scripting Interpreter	Post-compromise actions often require scripts, installers, or living-off-the-land execution.
Persistence	T1098 Account Manipulation	Account/token changes are common in identity-heavy intrusions.
Command and Control	T1105 Ingress Tool Transfer / T1090 Proxy	Payload retrieval or proxying may follow initial foothold.
Impact	T1486 Data Encrypted for Impact / T1041 Exfiltration Over C2 Channel	Ransomware/extortion and data theft remain common outcomes.

8. Infrastructure and Indicators

Hard IOCs were not consistently available across all public sources reviewed for this daily item. The practical observable set is behavioral.

Observable	Type	Operational Use
Unexpected package install or post-install script	Behavioral	Supply-chain compromise triage
Unusual direct-to-IP outbound traffic	Network behavior	Malware C2 hunting
New admin session from abnormal device/ASN	Identity behavior	Account takeover triage
RMM/tooling execution outside maintenance window	Endpoint behavior	Intrusion/ransomware precursor

9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Suspicious tool/package execution	DeviceProcessEvents / Sysmon	Find initial execution and staging
Direct-to-IP or rare outbound connections	DeviceNetworkEvents / firewall	Find C2 or staging without DNS
Identity and SaaS access anomalies	SigninLogs / audit logs	Find trusted-relationship abuse
title: NASA Core Flight System Vulnerability Daily Space Systems Review Daily Priority Signals id: nasa-core-flight-system-vulnerability-daily-space-systems-review-daily-20260805 status: experimental description: Flags endpoint or server-side activity aligned with NASA Core Flight System Vulnerability Daily Space Systems Review. Tune to environment before alerting. logsource: product: windows category: process_creation detection: selection_terms: CommandLine contains: 'cfs' CommandLine contains: 'core flight' CommandLine contains: 'housekeeping' CommandLine contains: 'telemetry'		

```

condition: selection_terms
fields:
  - Image
  - CommandLine
  - ParentImage
  - User
level: medium

```

10. Threat Hunting

10.1. Hunt Hypothesis

If this activity is present, analysts should find one or more of: suspicious execution tied to the topic keywords, outbound communications to rare infrastructure, or identity events inconsistent with the user's normal administrative pattern.

```

// Daily hunt: NASA Core Flight System Vulnerability Daily Space-Systems Review
let lookback = 14d;
DeviceProcessEvents
| where Timestamp > ago(lookback)
| where tolower(ProcessCommandLine) has "cfs" or tolower(RemoteUrl) has "cfs" or
tolower(ProcessCommandLine) has "core flight" or tolower(RemoteUrl) has "core flight" or
tolower(ProcessCommandLine) has "housekeeping" or tolower(RemoteUrl) has "housekeeping" or
tolower(ProcessCommandLine) has "telemetry" or tolower(RemoteUrl) has "telemetry" or
tolower(ProcessCommandLine) has "nasa" or tolower(RemoteUrl) has "nasa"
| project Timestamp, DeviceName, AccountName, FileName, ProcessCommandLine, InitiatingProcessFileName
| union (DeviceNetworkEvents
  | where Timestamp > ago(lookback)
  | where tolower(RemoteUrl) has "cfs" or tolower(RemoteUrl) has "core flight" or tolower(RemoteUrl)
has "housekeeping" or tolower(RemoteUrl) has "telemetry" or tolower(RemoteUrl) has "nasa"
  | project Timestamp, DeviceName, AccountName=InitiatingProcessAccountName,
FileName=InitiatingProcessFileName, ProcessCommandLine=strcat(RemoteUrl, ':', RemotePort),
InitiatingProcessFileName)
// SaaS and identity anomaly hunt for current daily threat review
SigninLogs
| where TimeGenerated > ago(14d)
| where ResultType == 0
| summarize Apps=dcount(AppDisplayName), IPs=dcount(IPAddress), Locations=dcount(Location) by
UserPrincipalName, bin(TimeGenerated, 1d)
| where Apps > 8 or IPs > 5 or Locations > 3
| order by TimeGenerated desc

```

11. Risk Assessment

Dimension	Rating	Rationale
Likelihood	Medium	Public reporting indicates an active or recently emphasized pattern.
Impact	High	Trusted access, ransomware, or supply-chain paths can create enterprise-wide exposure.
Detection Confidence	Medium	Behavioral detections are feasible but require tuning and baseline context.
Response Urgency	High	Same-day review prevents high-value signals from aging out of hot telemetry windows.

12. Recommended Actions

Priority	Owner	Action
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Critical	SOC	Run the included hunts and preserve positive-result telemetry.
High	Identity / SaaS	Review privileged access, recent token/session anomalies, and third-party accounts.
High	Endpoint / Platform	Confirm EDR coverage and script/package execution logging on exposed systems.
Medium	Governance	Add this item to the daily intel review queue and brief affected system owners.

13. References

- [1] CISA NASA cFS ICSA-26-211-06: <https://www.cisa.gov/news-events/ics-advisories/icsa-26-211-06>
- [2] CISA NASA cFS ICSA-26-197-03: <https://www.cisa.gov/news-events/ics-advisories/icsa-26-197-03>
- [3] NASA Core Flight System project: <https://cfs.gsfc.nasa.gov/>